Micrometrix®

PCA Model 247

Particle Charge Analyzer with Automatic Titration

Instruction Manual DRAFT

Document
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0.9.x

About this manual

This manual describes the setup, operation and routine maintenance of the Micrometrix Model PCA. It is written for laboratory and plant technicians who run charge titrations. Engineering details of the internal wiring are kept in a separate document, *PCA Wiring Reference*, and are not needed for day-to-day use.

Keep this manual with the instrument. If the instrument changes hands, pass the manual on with it.

Conventions used in this manual

WARNING	Indicates a situation that could result in injury, chemical exposure, or damage to property if the instruction is not followed.
CAUTION	Indicates a situation that could result in damage to the instrument, an incorrect result, or loss of data.
NOTICE	Information that helps you use the instrument correctly, with no safety implication.

On-screen buttons are shown as [ZERO]. Menu paths are written *Config* → *Titrate* → *Settle time*. Figures and tables are numbered by chapter, for example **Figure 5-1**. Volumes on the instrument are shown in microlitres (uL); 1000 uL = 1 mL.

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1 Safety information

Read this section before installing or operating the instrument.

WARNING	**Chemicals.** Titrants and samples may be hazardous. Read the safety data sheet for every reagent before use, wear the protective equipment it specifies, and follow your laboratory's procedures for spills and disposal.
WARNING	**Pump runs without supervision.** During an automatic titration, and while priming, the pump starts and stops on its own. Keep fingers, hair and loose clothing away from the pump head and tubing whenever the instrument is powered. Tap STOP on the Titrate screen during an automatic titration, or on the Pump screen (*Config* → *Pump* → *Pump controls*), to stop the pump immediately.
WARNING	**Electrical.** Installation and connection of power must be carried out by a qualified person in accordance with local regulations. Isolate the supply before opening the enclosure.
CAUTION	**Liquids and electronics.** Keep titrant bottles and sample vessels where a spill cannot reach the instrument's electronics or connectors. Check tubing connections before each session; a loose fitting can spray titrant when the pump runs.
CAUTION	**Continuous priming.** If *Prime auto-stop* is set to 0, a prime runs until it is stopped by hand. Do not leave the instrument while priming with this setting, or the titrant bottle can be emptied.
NOTICE	The instrument restarts itself automatically if its software ever stops responding, and the pump stops when it does. A titration in progress at that moment is lost. See section 11.

2 Product overview

2.1 Purpose

The Model PCA measures the **electrical charge carried by particles** in a liquid sample and reports it in millivolts on a colour touchscreen. A negative reading means the particles carry a net negative (anionic) charge; a positive reading, a net positive (cationic) charge.

The instrument can **titrate** the sample: a titrant of opposite charge is added in small, controlled doses by the built-in pump until the reading reaches zero, the charge-neutral point. **The volume of titrant needed is the result** — a direct measure of how much charge the sample carries. Every run is recorded second by second, kept on the instrument, and can be copied to a USB stick as a spreadsheet file.

2.2 Features

Function	Description
Measurement	Particle charge, −1000 to +1000 mV, from the sensor's 4–20 mA signal, with adjustable smoothing and sensitivity.
Sensor fault detection	A cut sensor cable or a failed sensor is shown as **SENSOR FAULT**, never as a reading, and ends a titration in progress.
Relative / absolute reading	Absolute is the sensor value; relative is measured from the last zero point. Either can be the main display.
Automatic titration	Adds titrant in shrinking steps that approach zero from one side, waits for the sample to settle after each, and stops at the endpoint.
Manual titration	Records a run while the operator adds titrant; stops automatically when the reading crosses zero.
Titrant pump	Stepper-driven peristaltic pump. Delivers single drops, set doses or continuous flow; calibrated in microlitres per motor step.
Titration curve	Live chart of the reading through the run, with the titrant total and dose count.
Signal Health	Live sensor waveform, amplitude, motor speed and a health score.
Run storage	Every run is kept on the instrument — about 600 runs of 5 minutes — with a results log. When it is full the oldest runs are removed.
Export	One CSV file per run to a USB stick: result, run conditions and the full curve. All stored runs can be copied at once.
Clock	Battery-backed real-time clock for run dates and file names.
Signal Health history	A record of sensor condition every 12 hours, kept for 60 days and viewable as a graph.
Calibration reminder	Flashes **CAL DUE** when the calibration interval has passed. Calibration settings are protected by a service PIN.

Table 2-1 Instrument functions

NOTICE	**From volume to charge demand.** The instrument reports titrant volume. To express the result as charge demand, multiply the titrant volume by the titrant concentration and divide by the sample volume. The titrant, its concentration and the sample volume are set by your test method.

3 Front panel and screens

All operation is through the touchscreen. Five buttons down the left edge select the main screens; the bar across the top shows the Micrometrix logo, the model, the current screen and the time of day.

TO DO — recapture the screenshots in this draft on firmware 0.9.20. They were taken on 0.9.17 and show the older top bar (an “M” mark, “PCA” and a cyan clock), a Pump button on the left rail instead of Files, four tiles on Signal Health, and the second reading in the footer rather than under the chip. The text describes 0.9.20.

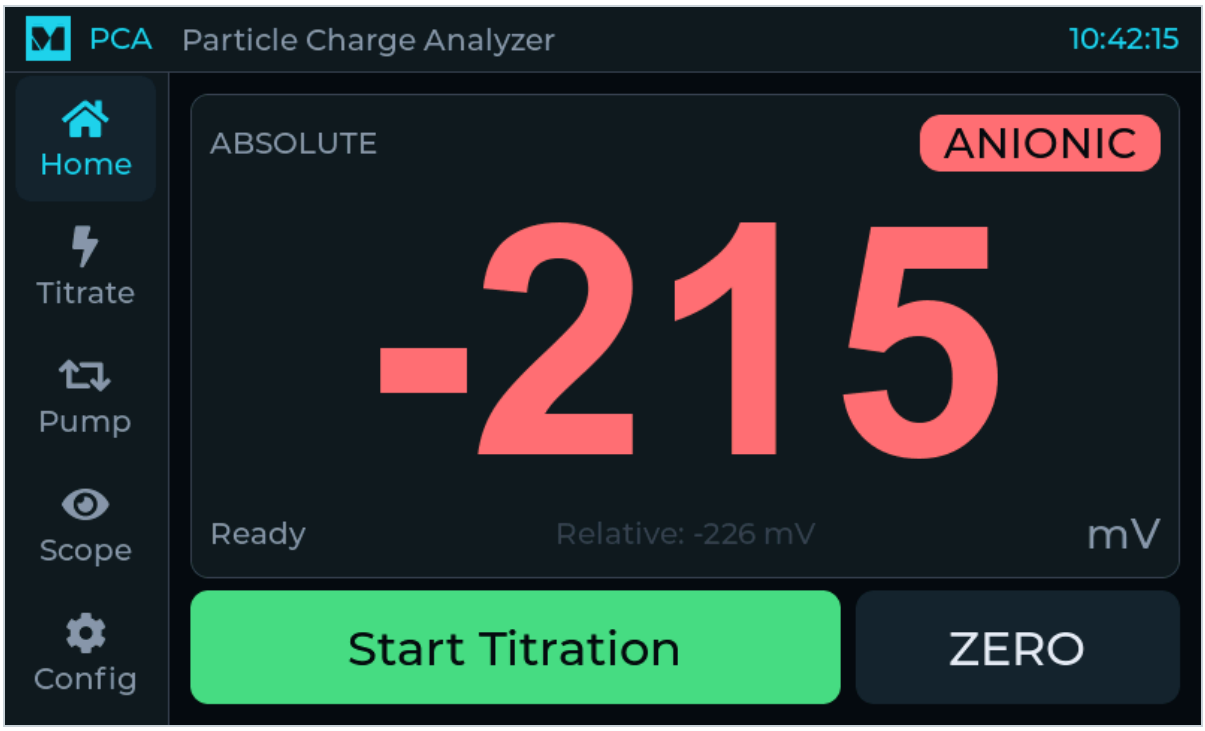


Figure 3-1 The Home screen, showing the absolute reading

RELATIVE / ABSOLUTE — which reading the main number shows: a yellow chip for relative, an outline for absolute. Tap it to switch. The small grey line under it is the other reading.

Main number — charge in mV; red when anionic, blue when cationic, cyan when neutral; **----** during a sensor fault.

Start Titration — start a manual titration (section 5.3). Reads **Stop Titration** during a run.

Charge chip — ANIONIC below −5 mV, NEUTRAL within ±5 mV, CATIONIC above +5 mV; **SENSOR FAULT** when the sensor input has failed (section 4.5).

Status — *Ready*; *Titrating m:ss* during a run; *Last test* with the date and time after one; *Check the sensor cable* during a sensor fault.

ZERO — make the current reading the zero point for the relative reading (section 4.2).

3.1 Screen summary

Button	Screen	Purpose
Home	Main reading	The reading, its charge sign, zero, and manual titration start/stop.
Titrate	Titration	Automatic titration, the live curve, titrant total and how the run ended.
Scope	Signal Health	Condition of the sensor itself: waveform, amplitude, motor speed, health score, and its history.
Files	File Utilities	The runs stored on the instrument and copying them to a USB stick; exporting and importing the configuration (chapter 8).
Config	Settings	Display, calibration, titration and pump settings, the clock, and the way to the Pump and Service screens.

Table 3-1 Main screens

Two further screens are opened from Config: the **Pump** screen (*Config → Pump → Pump controls*, chapter 6) and the **Service & Calibration** screen (*Config → Service*, section 13.1).

4 Daily operation

4.1 Reading the display

The main number is the particle charge in millivolts, updated four times a second. Its colour and the charge chip give the sign at a glance. Tap the **RELATIVE / ABSOLUTE** chip to switch the main number between the two readings (or use *Config → Display & clock → Main readout*). The other reading is always shown in small grey text under the chip, for example *Absolute −812 mV* while the main number is relative.

4.2 Zeroing

Zeroing sets the reference point for the relative reading. The titrator works on the **relative** reading and stops when it reaches zero, so the zero point defines the endpoint of every titration.

1. Allow the reading to settle.
2. Tap ZERO and confirm.
3. The relative reading becomes 0. The absolute reading is unchanged.

CAUTION	Do **not** zero on a sample you are about to titrate. Its reading becomes 0, and the titrator will report it as already at the endpoint. Zero only when your test method calls for it. Zeroing is not calibration: it moves the reference point only.

NOTICE	Zeroing, or changing calibration or titration settings, while a titration is running is allowed but is recorded in the exported file (`Settings_changed_mid_run,YES`).

4.3 Entering values

Tapping any setting opens a numeric keypad. Enter the value, use +/- for negative numbers, the backspace key to delete the last digit, Clear to start again and Enter to save. Cancel leaves the setting unchanged. The keypad title names the setting and the line beneath it gives the allowed range or a hint. A service PIN is shown as dots.

Figure 4-1 — TO DO: screenshot of the keypad open on a setting

4.4 CAL DUE reminder

The instrument keeps track of when it was last calibrated. When the calibration interval set by your service provider has passed, a red **CAL DUE** badge flashes in the top bar on every screen.

- The instrument **keeps working normally**: measurement and titration are not affected.
- Tap the badge to see when calibration fell due.
- Only your service provider can clear it, by recording the calibration on the Service & Calibration screen (section 13.1).

NOTICE	CAL DUE needs the clock. With no clock fitted or set, the instrument cannot tell how long it has been, and the reminder does not appear.

4.5 Sensor fault

The sensor sends its reading as a 4–20 mA current: 4 mA is −1000 mV and 20 mA is +1000 mV. A cut or unplugged cable, or a sensor without power, sends no current at all. The instrument recognises this and does not show it as a reading: below 3.6 mA for half a second, the main number shows **----**, the chip reads **SENSOR FAULT** in red and the status line reads *Check the sensor cable*.

- A titration in progress ends at once and the pump stops. The run is recorded as *Sensor fault – check the sensor cable* (Table 5-2) and is not a result.
- Start Titration, Auto Titrate and ZERO are dimmed and do nothing until the fault clears.
- The display returns by itself once the signal is back above 3.8 mA. Nothing needs to be reset.

CAUTION	Fault detection depends on the input calibration (*Config → Sensor*, section 9). If it has not been calibrated for the 4–20 mA signal, a cut cable may read as −1000 mV instead. Calibration is a service task.

5 Running a titration

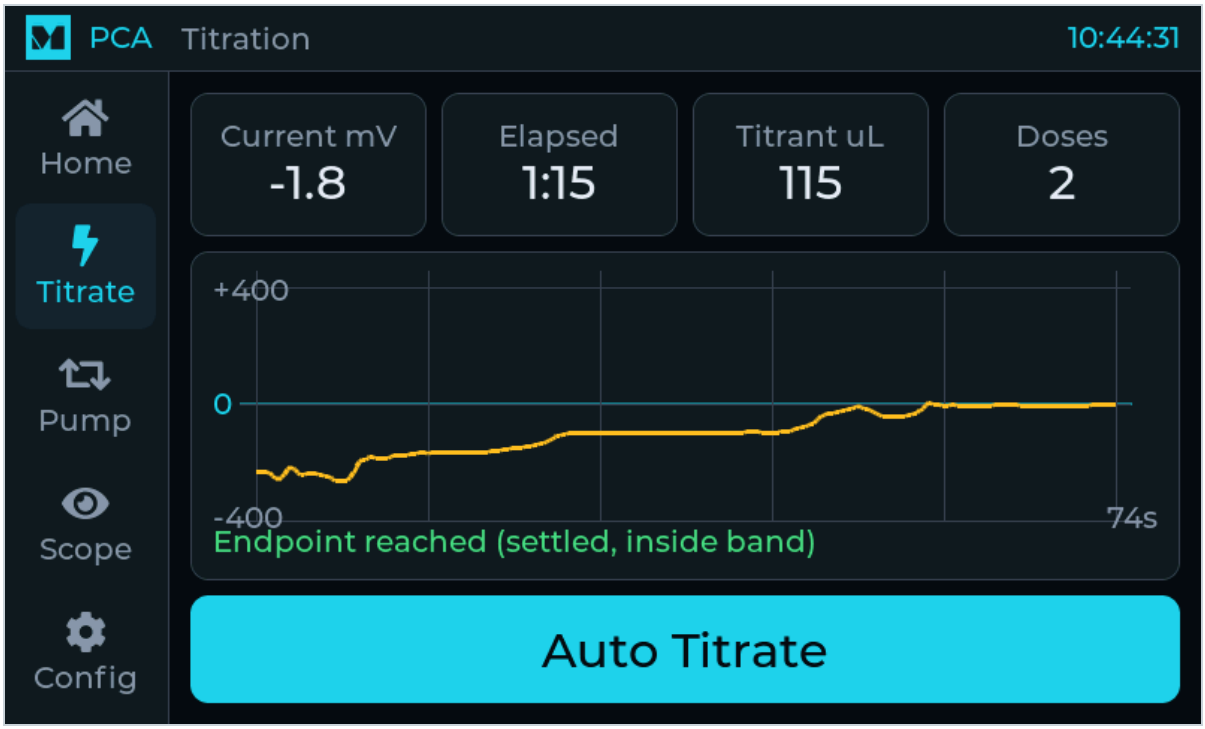


Figure 5-1 Titrate screen after an automatic titration has reached its endpoint

Current mV — the live relative reading.

Titrant uL — titrant delivered by the pump in this run. **This is the result.**

Chart — the reading through the run, drawn live: the end of the line is the reading now. The cyan line is zero and the green band around it is the endpoint band; the run is finished when the line settles inside it. While a run is in progress the scale zooms in on the present reading as it approaches zero; the label at right is the run length.

Auto Titrate — start an automatic titration, after a confirmation. Reads **STOP** while one is running.

Elapsed — time since the run started.

Doses — number of doses the automatic titrator has made.

Status line — what the titrator is doing now, then how the run ended (section 5.5).

5.1 Before you start

- The titrant line is primed: titrant reaches the tip with no air in the tubing (section 6.2).
- The pump is calibrated (section 6.3). An uncalibrated pump gives a wrong volume even when the endpoint is found correctly.
- The sample is prepared in the measuring cell as your test method specifies, and the reading on Home is steady.
- Nothing to prepare for the record: every run is saved on the instrument when it ends (chapter 8).

TO DO: sample volume, measuring cell preparation, stirring, and titrant recommendations for the finished product.

5.2 Automatic titration

1. Open the **Titrate** screen. Check the **Mode** beside Auto Titrate (tap it to change; see below), then tap Auto Titrate. A dialog states the mode, the opening dose and the most titrant the run will add (*Max titrant*). Tap Confirm to start, or Cancel.
2. The status line reads *Waiting for a stable baseline…* until the reading holds steady.
3. The instrument adds an **opening dose** (*Opening dose* × *uL per drop*; 5 drops by default).
4. After each dose it waits the *Settle time*, then waits again until the reading holds steady for 2 seconds.
5. From how far the doses have moved the reading, it predicts the volume still needed to reach zero and adds a fraction of it (the *Approach factor*, 60% by default), never more than the *Largest dose*. Doses therefore shrink as the reading nears zero and approach it from one side, ending on single drops.
6. Each dose is at most one and a half times the one before, so the doses grow step by step instead of jumping to the *Largest dose*. When one dose suddenly moves the reading much more than the ones before it — the steep part of the titration curve, typical of polyelectrolytes such as DADMAC — the instrument predicts from that dose alone and limits every later dose to a quarter of it. The status line shows *ramp* or *steep* after the dose when either applies.
7. The run ends by itself (section 5.5). Read the result from the **Titrant uL** tile.
8. The run is saved on the instrument automatically (chapter 8).

Mode. The titrator does not plan the whole run in advance: after every dose it measures the response and decides the next dose. The mode sets how boldly it does that — three settings at once:

Mode	Approach factor	Largest dose	Settle time	Use for
Cautious	40%	10 drops	15 s	Slow or unfamiliar samples, poor mixing, or when overshoot must be avoided. About twice the doses of Normal and the longest runs.
Normal	60%	20 drops	10 s	Routine samples. The factory setting.
Aggressive	85%	40 drops	6 s	Well-mixed samples that respond quickly. Fewest doses and shortest runs; the most likely to cross zero if the sample responds slowly.

Table 5-2 Titration modes

The mode changes how the titrator reaches the endpoint, not where the endpoint is: *Endpoint band*, *Opening dose* and *Max titrant* are the same in every mode, so results from different modes compare directly. The three values can also be set one by one in *Config* → *Titrate*; if they then match no mode, the mode reads **Custom**. The mode cannot be changed while the titrator runs. Each run's file records it (`AT_mode`).

To stop early, tap STOP. It does not ask: the pump stops at once and the run is recorded as stopped by the operator.

Status line during a run	Meaning
Waiting for a stable baseline…	Checking the reading is steady before the first dose.
Dose 3: 38 uL (1462 motor steps)	Dose number, volume, and the motor steps commanded for it.
… **min** at the end	The calculated dose was smaller than one drop, so one drop was used.
Dose 3: settling…	Waiting for the sample to respond to the last dose.
(slope … mV/mL)	How strongly the reading responds to titrant; used to predict the next dose.
Dose 3: reading moving AWAY from zero	The last dose moved the reading the wrong way. See section 11.

Table 5-1 Titrator messages

CAUTION

Overshoot cannot be undone. Titrant cannot be removed from a sample. If runs often end with *Endpoint (crossed zero)*, the sample is being dosed faster than it responds: choose a more cautious **Mode**, or increase *Settle time* or reduce *Approach factor* (chapter 9).

5.3 Manual titration

1. On **Home**, tap Start Titration. The status reads *Titrating* with the elapsed time.
2. Add titrant with the 1 Drop or Dose buttons on the **Pump** screen (*Config* → *Pump* → *Pump controls*). The reading is recorded every second and the titrant is counted.
3. When the reading changes sign, the instrument records two more seconds and ends the run. Or tap Stop Titration at any time.

CAUTION

Only titrant delivered by the instrument's pump is counted. Titrant added by pipette is not included in the result.

NOTICE

You can tap Auto Titrate during a manual run. After you confirm, the titrant already added is kept in the total, counts toward *Max titrant*, and the automatic titrator continues from there.

5.4 Long runs

The curve holds 600 points, one per second: ten minutes. A run that lasts longer continues and the titrant total keeps counting, but the chart stops extending and the exported file is marked `Curve_truncated,YES`.

5.5 How a run ends

Status line	Result	Meaning
Endpoint reached (settled, inside band)	Valid	The reading settled within the *Endpoint band* of zero. The normal ending.
Endpoint – within one drop	Valid	Less than one drop was still needed. Adding it would have overshot, so the run stopped.
Endpoint (crossed zero)	Valid	The last dose carried the reading past zero. The result is slightly high, by at most that dose.
Already at endpoint – nothing to titrate	Valid	The sample was within the endpoint band before any titrant was added.
Passive zero crossing (unsettled)	Valid	Manual run: the reading changed sign.
Max volume reached	Incomplete	The next dose would exceed *Max titrant*. The true result is higher than the volume shown.
Too many steps	Incomplete	200 doses were made without reaching the endpoint.
No response – check for stall, empty titrant or air in line	Fault	Eight doses in a row, and at least a tenth of *Max titrant*, together moved the reading less than 3 mV. See section 11.
No slope – dose produced no change	Fault	The reading did not respond to the opening dose. See section 11.
Opening dose exceeds max titrant	Fault	Settings conflict: the opening dose alone is larger than *Max titrant*. No titrant was added.
Sensor fault – check the sensor cable	Fault	The sensor signal failed during the run (section 4.5). The pump stopped. Automatic or manual runs.
Stopped by operator	Stopped	STOP or Stop Titration was tapped.

Table 5-2 Run endings

6 Titrant pump and calibration

The Pump screen is used for priming, for adding titrant during a manual titration, and when setting up: measuring drop size and calibrating. Open it from *Config* → *Pump* → *Pump controls*, or from the Service & Calibration screen.

NOTICE	**Calibration values are protected.** *Drop uL*, *uL/step* and [Enter mL] can only be used with the service PIN. Tapping one while it is locked asks for the PIN; after a correct PIN the keypad for that value opens. Priming, drops, doses and [Cal Run] do not need the PIN.

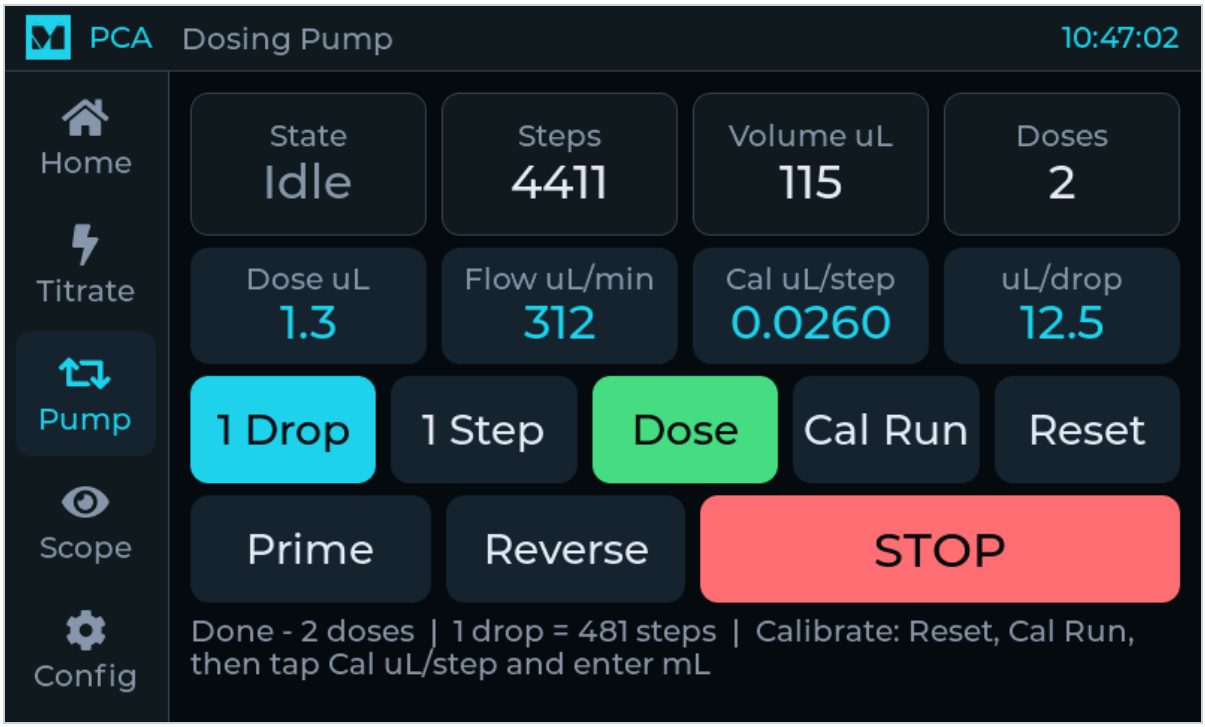


Figure 6-1 Pump screen (taken on 0.9.17, before the screen was arranged in sections)

6.1 Controls

The screen has a readout line at the top, then three sections. Each section has its own speed: Prime, Dose and Calibrate each run at the flow set in that section, so changing one does not change the others. Each section is named in blue with a line saying what it is for, and holds its buttons followed by the settings those buttons use. Tap a setting to change it on the keypad. [STOP], bottom right, stops whatever the pump is doing.

Section	Control	Function
—	State, Steps, Volume uL, Doses	What the pump is doing, and the motor steps, volume and doses delivered since [Reset].
Prime	[Prime]	Run forward until stopped, to fill the tube. Reads **Stop** while the pump is running.
	[Reverse]	Run backward until stopped, drawing titrant back toward the bottle.
	Flow uL/min	The rate for Prime and Reverse.
	Auto-stop s	A prime stops itself after this time; *Never* = run until stopped.
Dose	[1 Drop]	Deliver one drop (*Drop uL*).
	[Dose]	Deliver one dose (*Dose uL*).
	[1 Step]	Turn the motor one step, a few nanolitres. Also a wiring check for service.
	Drop uL	Volume of one drop from the tip (section 6.4). Service PIN.
	Dose uL	Size of one dose.
	Rate uL/min	The rate for 1 Drop, Dose and 1 Step. Slower forms cleaner drops. It does not change the prime or Cal Run rate.
Calibrate	[Reset]	Set Steps, Volume and Doses back to zero.
	[Cal Run]	Run a fixed 50,000 motor steps into a weighed container.
	Flow uL/min	The rate for Cal Run (default 1500 motor steps a second, about 2320 uL/min with the factory calibration; about 33 s for a run).
	[Enter mL]	Enter the millilitres collected; the instrument divides by the steps since Reset (section 6.3). Service PIN.
	uL/step	The calibration: microlitres per motor step. Tap to type it directly if already known. Service PIN.

Table 6-1 Pump screen controls

The line beside [STOP] shows what the pump is doing or last did, and, when idle, the number of motor steps in one drop.

NOTICE	**Volumes are counted, not measured.** The instrument counts the motor steps it sends and converts them with the calibration. Without *Power sense* (section 6.5) it cannot tell whether the pump is connected and powered: with the motor supply off, the counts still rise while nothing is dispensed.

6.2 Priming

1. Place the titrant tube in the titrant bottle and the tip over a waste container.
2. Tap [Prime]. Let titrant run until it leaves the tip in a steady stream with no air bubbles in the tubing.
3. Tap [STOP] (or [Stop]).

Priming runs at the **Flow uL/min** in the Prime section, not at the Dose section's **Rate uL/min**. The status line gives the rate while priming, and a rate changed while the pump runs takes effect at once. The fastest the pump runs is 4000 motor steps a second, about 6 mL/min with the factory calibration; a higher value is reduced to that and the status line says so. Prime before every session and after changing the titrant bottle.

6.3 Calibrating the pump

Calibrate when the instrument is installed, after replacing pump tubing, and whenever titrant volumes appear wrong. A balance reading to 0.001 g is recommended. With water, 1 g is 1 mL.

1. Prime the line (section 6.2).
2. Place a tared container under the tip and tap Reset.
3. Tap Cal Run and wait for the readout to show **Idle**. It runs at the Calibrate section's *Flow uL/min*; a rate close to the dosing rate measures the calibration as doses will deliver it, at the cost of a longer run.
4. Weigh the liquid collected.
5. Tap Enter mL (enter the service PIN if asked), enter the **millilitres collected** and tap Enter. The status line shows the old and new calibration and the step count used.
6. Repeat steps 2–5. The two results should agree within a few percent.

CAUTION	Do not tap Reset between Cal Run and entering the volume. The calibration divides by the steps counted since Reset; the keypad states that count, and refuses if it is zero.

6.4 Measuring drop size

Drop size depends on the tip and the titrant, so measure it with the titrant you will use.

1. Prime the line, then set a slow prime **Flow uL/min** so drops can be counted.
2. Tap Reset, then Prime, and count a number of drops (50 or more).
3. Tap STOP and read the **Volume uL** tile.
4. Divide that volume by the number of drops and enter the result in **Drop uL** (service PIN).

The titrator never doses less than one drop, so a smaller drop allows a finer approach to the endpoint.

6.5 Pump power sense

If the pump's motor supply is wired to the instrument's sense input (*PCA Wiring Reference*), turn on *Config → Pump → Power sense (D5)*. The row then shows **On – powered** or **On – NO POWER**. While there is no motor supply:

- the Pump screen shows **NO POWER** and *No pump power – check the motor supply. Nothing can be dispensed.*, and its buttons do nothing;
- Auto Titrate is dimmed and will not start;
- if the supply is lost during a run, the pump stops and an automatic titration ends with *No pump power – check the pump supply*. The run is saved with the outcome `ABORT_PUMP_POWER` and the status `FAULT`: it is not a result. A manual titration carries on.

Leave *Power sense* off on an instrument without the sense wiring: it would report NO POWER all the time. The sense sees the supply, not the motor itself, so a motor cable pulled out with the supply on is not detected; the titrator's *No response* check (chapter 11) is the backstop for that.

7 Signal Health

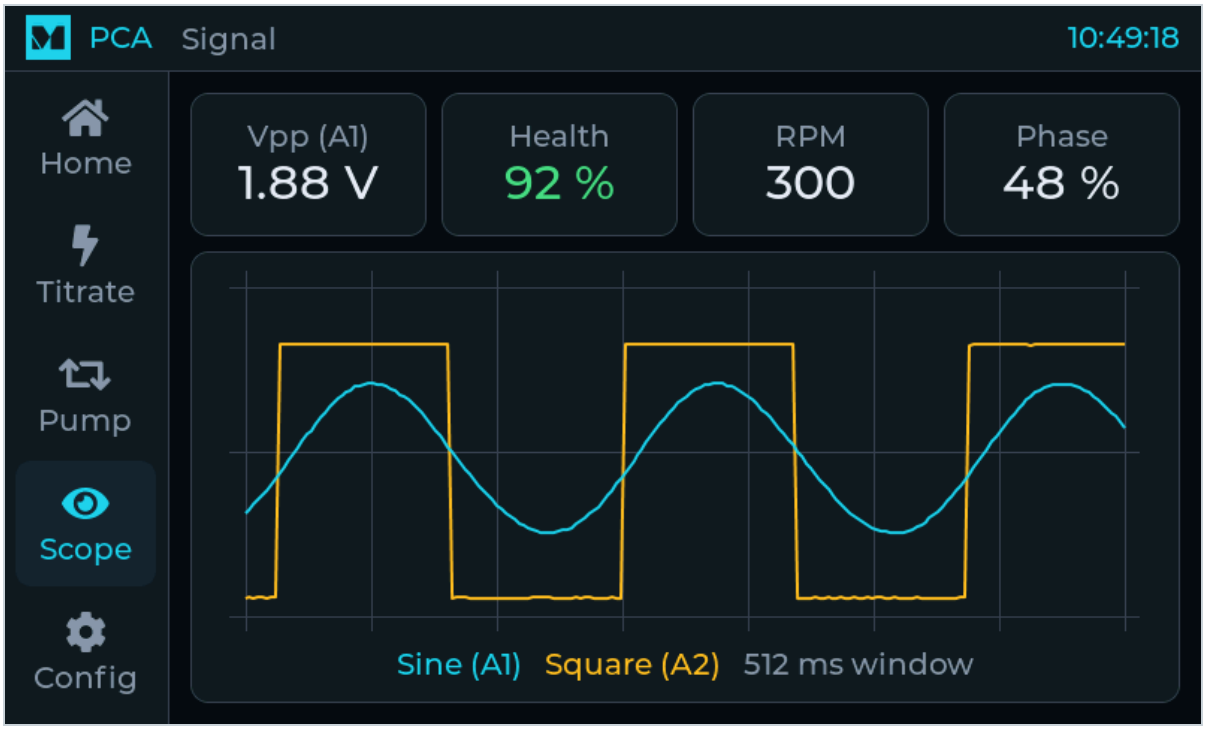


Figure 7-1 Signal Health. Sensor waveform (blue) against the drive reference (amber), 512 ms window. (Taken on 0.9.17, which still showed four tiles.)

This screen shows the sensor itself. Both traces are centred on the middle line, as on an AC-coupled oscilloscope.

Tile	Meaning
Sine Vpp	Peak-to-peak amplitude of the sensor signal, in volts, at the instrument input.
Signal Health	0–100% score of waveform quality: how clean and sine-like the signal is, and whether it is correctly timed against the drive. Green 85% and above, amber 60–84%, red below 60%. Tap the tile for its history (section 7.1).
Motor RPM	Sensor motor speed, from the drive reference.

Table 7-1 Signal Health tiles

A healthy sensor gives a clean, regular blue waveform locked to the amber reference. Ragged traces, a falling health value, or a falling Vpp indicate a fouled or worn sensor, a stalled motor or a wiring fault — usually before the reading itself is obviously wrong. With no usable signal the plot shows **No sensor signal** and the Health and RPM tiles read `--`; the small wiggle still visible then is electrical noise on an unconnected input, not a signal.

7.1 Signal Health history

Figure 7-2 — TO DO: screenshot of the Signal Health history, 30-day view

Tap the **Signal Health** tile to see how the health of the sensor has changed over the last `7 days`, `30 days` or `60 days`. A slow decline is the typical sign of a sensor that is fouling; a sudden step down points to an event at that

time. Close the graph with [×].

Item	Meaning
Blue line	Average health over each 12-hour period (00:00–12:00 and 12:00–24:00).
Grey line	Lowest 1-minute average in the same period. A grey line that falls away from the blue one means short drops rather than a general decline.
Right-most point	The period in progress, as it stands when the graph is opened.
Gaps	No record for that period: the instrument was switched off, or the sensor signal could not be read.
Line under the graph	The period in progress: hours so far, average and lowest health, share of the time the signal was readable, motor speed and its spread, sine amplitude.

Table 7-2 Signal Health history

The sensor is checked continuously, whichever screen is showing. One record is kept for each 12-hour period, for 60 days; after that the oldest is replaced. Records survive a power interruption. The history needs the clock to be set, and a period in which the instrument ran for less than 30 minutes is not recorded. The scale runs from 80 to 100% and extends downward in steps of 20% when a value is lower.

8 Stored runs and export

Every titration run is saved on the instrument the moment it ends: the complete file described in section 8.2, and a line in the results log (section 8.3). The instrument keeps about 600 runs of 5 minutes, or 300 of the 10-minute maximum. When it is full, the oldest run is removed to make room, so saving never stops. A USB stick is needed only to copy runs to a computer, and it is written only when you export: a stick left in the instrument is not written to. Wait for *Safe to remove the stick* before pulling it out.

CAUTION	A run is saved when it ends. A power interruption during a run loses that run; runs already finished are not affected.

8.1 File Utilities

Figure 8-1 — TO DO: screenshot of the File Utilities screen after an export

Open the screen with the **Files** button on the left. It opens on a list of three: **Titration runs**, **Export configuration** and **Import configuration** (section 8.4). Tap **Titration runs**; ‹ Back, or **Files** on the left again, returns to the list. The page shows the **Last run** (its date and start time), its **Duration**, the file name it was **Exported as**, and what is **Stored on unit** — the number of runs and the space used, for example *214 runs, 2.6 MB of 7 MB* — followed by the state of the USB stick. Fit a stick and wait until it is reported ready, then:

- Export run (green) copies the last run. The status line confirms *Saved …* with the file name and *Safe to remove the stick*, and **Exported as** shows it in the same green.
- Export all (violet) copies every run stored on the instrument that is not on the stick yet, with the results log (`PCA_results.csv`) and the service log (`PCA_service_log.csv`). It asks first. The copy runs in the background: a bar shows the progress and the status line counts the runs (*Copying runs… 37 of 212*) in blue, then gives the number copied and *Safe to remove the stick*. The screen stays usable, but a titration cannot be started until the copy is finished. Leave the stick in until then.

Files are named from the run's start date and time and its run number, for example `2026-09-17_10-43-17_012.csv`. If the clock is not running the name is `PCA_run_012.csv`, and the screen shows *no clock* in place of the date. Exporting the same run again replaces the earlier file of the same name.

NOTICE	A run can be exported while it is still in progress. The file is then marked `Outcome,IN_PROGRESS` and is not a result.

8.2 File format

Plain text, comma separated. A header of `Name,Value` lines describes the run, then a blank line, then one row per second. Example with illustrative values, shortened:

```
Micrometrix Model PCA - Titration Export
Schema,4
Firmware,0.9.33
Run_id,12
Outcome,ENDPOINT_BAND
Outcome_detail,Endpoint reached (settled, inside band)
Result_status,OK
Training_use,CURVE
Mode,AUTO
Date,2026-09-17
Start_time,10:43:17
Duration_s,74
Samples,75
Clock,RTC
Clock_trusted,YES
Total_titrant_uL,115.0
Total_titrant_mL,0.1150
Auto_doses,2
...

Second,mV,Titrant_uL,Titrant_mL,Phase,Dose,Settled,Clipped,Manual
0,-212.4,0.0,0.0000,BASE,0,0,0,0
1,-212.1,0.0,0.0000,BASE,0,1,0,0
...
[END],75
```

Header line	Meaning
Result_status	**OK** a valid result; **INCOMPLETE** the endpoint was not reached; **FAULT** the run stopped on a fault; **STOPPED** stopped by the operator; **NOT_A_RESULT** exported while running.
Outcome / Outcome_detail	How the run ended, as a code and as the message shown on screen (Table 5-2).
Mode	AUTO if the automatic titrator made any doses, otherwise MANUAL.
Total_titrant_uL / _mL	The result: titrant delivered in the run.
Volume_is_lower_bound	YES when the run stopped at a volume or step limit; the true result is higher.
Clock_trusted	NO if the date and time did not come from a running clock.
AT_…, Pump_…, Gain, Avg_samples, Zero_offset_mV	The titration, pump and calibration settings in force, so a result can be checked or corrected later.
Settings_changed_mid_run	YES if zero, calibration or titration settings were changed during the run.
Other lines	Diagnostic information about signal noise, doses and pump condition, for service use. `Training_use` classifies the run for data analysis and can be ignored.

Table 8-1 Principal header lines

Column	Meaning
Second	Seconds since the start of the run.
mV	Relative reading.
Titrant_uL / Titrant_mL	Titrant delivered so far.
Phase	Titrator state: BASE baseline, DOSE dosing, SETL settling, DONE finished, IDLE manual run.
Dose	1 in the second a dose was delivered.
Settled	1 when the reading was steady.
Clipped	1 when the reading was at the ±1000 mV limit.
Manual	1 when titrant was added from the Pump screen rather than by the titrator.

Table 8-2 Data columns. The file ends with `[END]` and the row count; a file without it is incomplete.

CAUTION

USB media. Use the stick supplied with the instrument or one formatted FAT32. Sticks larger than 32 GB are often formatted exFAT and are not recognised. Some sticks can be read but not written by this instrument; if an export reports a write error, use a different, smaller stick. A failed export never leaves a partial `.csv` file on the stick.

8.3 Results log

The instrument keeps one line per run in a results log, copied to the stick as `PCA_results.csv` by Export all. Open it to follow results over weeks without opening each run's file.

```
Date,Start_time,Run_id,Mode,Outcome,Result_status,Titrant_uL,Titrant_mL,Duration_s,Auto_doses,File
2026-09-24,10:43:17,123,AUTO,ENDPOINT_BAND,OK,1129.2,1.1292,142,7,2026-09-24_10-43-17_123.csv
2026-09-24,11:02:40,124,MANUAL,ZEROCROSS_PASSIVE,OK,845.0,0.8450,301,0,2026-09-24_11-02-40_124.csv
```

The columns mean the same as the header lines of a run file (Table 8-1); **File** names the run's own file. The results log is not shortened when old runs are removed.

8.4 Configuration file

The settings can be saved to a USB stick as a plain text file, `PCA_Config.txt`, and loaded back into this or another PCA. Use it to set up several instruments the same way, or to keep a record of a setup.

- **Files → Export configuration** writes the file. The line under the list confirms *Saved PCA_Config.txt*. Exporting again replaces the file.
- **Files → Import configuration** asks first, then reads the file and replaces the settings. The line under the list gives the number of settings imported and of lines skipped. It is refused during a titration and while the pump is running.

The file has one setting per line, `name = value`, and can be read and edited in any text editor. Lines starting with `#` are comments.

```
# MicroMetrix PCA Model 247 configuration
# GigaView firmware 0.9.33, exported 2026-09-30 09:29:20
gain = 1.000
avg_samples = 32
main_readout_absolute = 0
at_open_drops = 5
at_max_ml = 10.000
...
# Calibration of the unit this file came from. For the record only:
# these lines are NOT read back by Import.
zero_offset_mv = 0.000
input_volts_at_4ma = 0.478
...
```

Import reads the operating settings: sensitivity, averaging, main readout, the five titration settings, pump, prime and Cal Run rates, dose size, prime auto-stop, brightness and the calibration reminder interval. Each value is checked against the range in Table 9-1; a line with an unknown name or a value out of range is skipped and nothing is changed for it. The last block of the file — the zero, the 4–20 mA input calibration, *uL per step* and *uL per drop* — belongs to the instrument it was exported from and is **never** imported. The service PIN is not written to the file.

NOTICE	The pump and prime rates are stored as motor steps per second. On an instrument whose *uL per step* differs, the same steps give a different flow in uL/min.

9 Settings reference

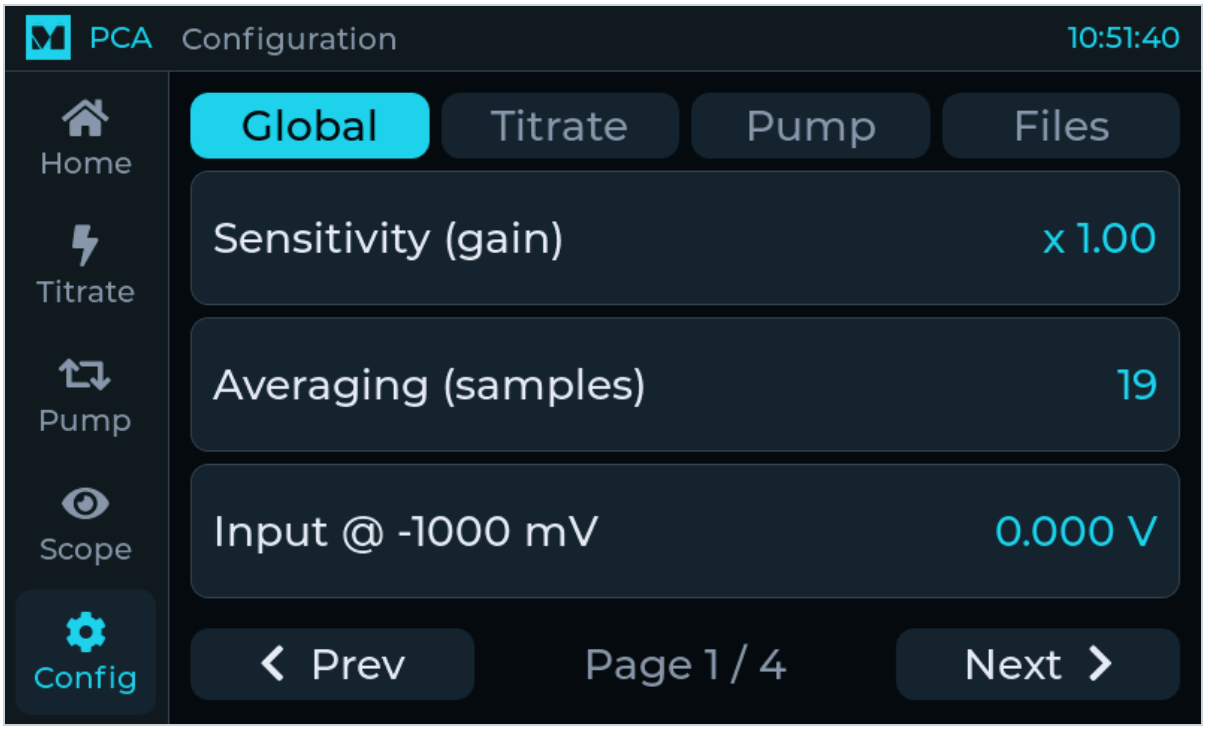


Figure 9-1 Settings. (Taken on 0.9.17, which still had tabs across the top.)

Config opens on a list of five groups. Tap a group to see its settings; ‹ Back, or **Config** on the left again, returns to the list, which is also where Config always opens. A group with more settings than fit scrolls with a finger. Within a group the settings used day to day come first, calibration and tuning after them; settings marked *Service PIN* ask for the PIN before they open. The last line of the list is the firmware version.

Group	Setting	Default · range	Function
Sensor	Sensitivity (gain)	×1.00 · 0.10–10.0	Multiplies the reading.
	Averaging (samples)	32 · 1–1000	Smoothing. Higher is steadier and slower.
	Input @ 4 mA (−1000 mV)	0.478 V · 0–3.3	Two-point calibration of the 4–20 mA sensor input: the input voltage with 4 mA and with 20 mA applied. The keypad shows the live voltage. Installation setting. Service PIN.
	Input @ 20 mA (+1000 mV)	2.340 V · 0–3.3	
	Raw ADC readout (diag)	Off	Adds the raw input value to the Home screen. Service use.
Titrate	Mode	Normal	Cautious, Normal or Aggressive: sets Approach factor, Largest dose and Settle time together (Table 5-2). Tap to change; also beside Auto Titrate.
	Opening dose	5 drops · 1–100	Size of the first dose, in drops.
	Max titrant	10.0 mL · 0.1–500	The titrator will not exceed this volume in one run.
	Endpoint band	2.5 mV · 0.5–100	How close to zero counts as the endpoint.
	Approach factor	60% · 10–100%	Share of the predicted remaining volume added per dose. Lower is slower and less likely to overshoot.
	Largest dose	20 drops · 1–100	The most one dose may add, however far the reading is from zero.
	Settle time	10 s · 1–600	Minimum wait after each dose before the reading is judged.
Pump	Pump controls	—	Opens the Pump screen, where every pump setting is made: the three flow rates, prime auto-stop, drop and dose size, and the calibration (chapter 6).
	Power sense (D5)	Off	On when the motor supply is wired to the sense input; the pump then refuses to run without its supply (section 6.5). Installation setting.
Display & clock	Main readout	Relative	Relative or Absolute as the main number. Tapping the RELATIVE / ABSOLUTE chip on Home does the same.
	Brightness	80% · 10–100%	Slider. Never fully dark. Kept through power cycles and firmware updates.
	Date	—	Enter as YYYYMMDD.
	Time	—	Enter as HHMM, 24-hour.
Service	—	—	Opens the Service & Calibration screen (section 13.1).

Table 9-1 Settings by group. Drop uL, Dose uL and Rate uL/min are set on the Pump screen, and the pump is calibrated there (Table 6-1).

NOTICE	Settings are stored in the instrument and kept through power loss and firmware updates. A firmware update does not reset calibration. A factory reset is on the Service & Calibration screen, behind the service PIN (section 13.1).

10 Installation and connections

WARNING Connections must be made by a qualified person with the supply isolated. See chapter 1.

Figure 10-1 — TO DO: end-user connection diagram of the enclosure (sensor, titrant pump and tubing, USB, power). Pin-level detail remains in the *PCA Wiring Reference*.

Connection	Notes
Particle charge sensor	Supplied sensor and cable.
Titrant pump tubing	Inlet from the titrant bottle; outlet tip over the measuring cell. *TO DO — tubing type and replacement interval.*
USB port	Export stick.
Power	*TO DO — specification of the finished enclosure supply.*

Table 10-1 External connections

10.1 First use

1. Set the date and time (*Config → Display & clock → Date / Time*).
2. Prime the titrant line (section 6.2).
3. Calibrate the pump (section 6.3) and measure the drop size (section 6.4). Both need the service PIN.
4. Record the calibration on the Service & Calibration screen (section 13.1), which starts the CAL DUE countdown.
5. Check Signal Health (chapter 7) with the sensor running.

11 Troubleshooting

Symptom or message	Likely cause and action
Main number shows **----** and **SENSOR FAULT**	The sensor signal is below 3.6 mA: the sensor cable is cut, unplugged or loose, or the sensor has no power. Check the cable and its connectors. The reading returns by itself when the signal does (section 4.5).
Sensor fault – check the sensor cable at the end of a run	The sensor signal failed during the run, so the run was stopped. Fix the cause above and run the sample again.
Pump screen shows **NO POWER**; *No pump power – check the pump supply*	The motor supply is off or unplugged. Switch it on and check its connector; the pump is available again within a second. If the supply is on, check the sense wiring, or turn *Power sense* off if it is not fitted (section 6.5).
Pump counts steps but nothing comes out	The motor supply is off, or the motor or tubing is not connected. With *Power sense* on, a missing supply is reported (section 6.5).
Calibrate the pump first	uL per step or uL per drop is zero. Calibrate (section 6.3) and set the drop size (section 6.4).
No response – check for stall, empty titrant or air in line	Eight doses in a row, and at least a tenth of *Max titrant*, together moved the reading less than 3 mV. Usually no titrant is reaching the sample: check the titrant bottle is not empty, prime to clear air, check the tubing is not pinched or disconnected, and that the pump turns. With a weak titrant the sample may simply respond little per dose: raise *Largest dose* or choose the **Aggressive** mode, so each dose is bigger.
No slope – dose produced no change	The opening dose was too small to move the reading, or did not reach the sample. Check the tip is over the cell; increase *Opening dose*.
Dose N: reading moving AWAY from zero	Titrant is moving the reading the wrong way: it may carry the same charge as the sample, or be the wrong titrant. Tap STOP and check the titrant. The titrator continues with single drops until stopped or until a limit is reached.
Max volume reached	The sample needs more titrant than *Max titrant*. Use a more concentrated titrant or a smaller sample as your method allows, or raise *Max titrant*.
Opening dose exceeds max titrant	Reduce *Opening dose* or *uL per drop*, or raise *Max titrant*.
Runs often end *Endpoint (crossed zero)*	Dosing is faster than the sample responds. Increase *Settle time*, reduce *Approach factor*, check stirring, or use a smaller tip.
Titrant volumes wrong or inconsistent	Recalibrate the pump (section 6.3). Check for air in the line and for worn tubing.
Reading noisy	Increase *Averaging*. Check Signal Health (chapter 7).
Signal Health shows **No sensor signal**	The sensor signal cannot be read. Check the sensor cable and that the sensor motor runs.
Signal Health history: *Clock not set* or *No readable signal yet*	Set the date and time (*Config → Display & clock*). The first point appears at the next 00:00 or 12:00 after at least 30 minutes of measuring.

Table 11-1 Measurement and titration

Symptom or message	Likely cause and action
CAL DUE flashing in the top bar	The calibration interval has passed. The instrument keeps working. Contact your service provider (section 4.4).
Keypad asks for a *Service PIN*	The setting is a calibration value and is protected. Only a service provider should change it.
Wrong PIN / *Too many attempts – wait 60 s*	Five wrong PINs lock the keypad for a minute. A forgotten PIN can be reset by a service provider.
No USB drive detected or *drive not readable (needs FAT32)*	Insert a FAT32 stick and wait a few seconds for it to be recognised. See the USB caution in chapter 8.
Can't write to stick (errno …) or *Export FAILED … not on the stick*	The stick can be read but not written. Use a different stick and export again; the run is still on the instrument.
Stored on unit shows *not available*	The instrument's run store could not be started. Runs are kept only until the next run or a power interruption. With no run store, a fitted USB stick receives each run when it ends; keep one fitted and contact your service provider.
Stored on unit shows *last save FAILED*	The last run could not be saved on the instrument. Export it now with Export run. If it happens again, contact your service provider.
Rename failed – data in ….csv.part	The run was written and checked, but the file could not be given its final name. The data is complete in the `.csv.part` file: rename it to `.csv` on a computer, or export again.
Clock shows `--:--:--` or blinks red	Clock module not responding. Dashes: not seen since power-up. Blinking: was working and has stopped. Files are then named `PCA_run_NNN`. Contact support.
Instrument restarted by itself	The instrument restarts if its software stops responding, and the pump stops. A run in progress is lost. If it recurs, note the time and contact support.

Table 11-2 Instrument, USB stick and clock

12 Specifications

Item	Value
Measurement range	−1000 to +1000 mV
Sensor input	4–20 mA (4 mA = −1000 mV, 20 mA = +1000 mV); below 3.6 mA reported as a sensor fault
Display update	4 per second
Display	Colour touchscreen, 800 × 480
Titration recording	1 sample per second; curve memory 600 samples (10 minutes)
Titrant pump	Stepper-driven peristaltic; volume per motor step set by calibration
Minimum titrator dose	One drop (set by the tip and titrant)
Maximum titrant per run	Adjustable, 0.1–500 mL
Endpoint detection	Settled within an adjustable band of zero (0.5–100 mV), or zero crossing
Export	CSV, one file per run, and a results log, to a FAT32 USB stick
Clock	Battery-backed real-time clock
Settings retention	Non-volatile; retained through power loss and firmware updates
Run storage	Every finished run, kept through power loss: about 600 runs of 5 minutes; the oldest removed when full
Signal Health history	One record per 12 hours, 60 days, retained through power loss
Power	*TO DO*
Enclosure / environment	*TO DO*

Table 12-1 Specifications

13 Service and support

For technical support, spare parts and calibration services, contact Micrometrix.

TO DO — company address, telephone, e-mail, web address; warranty statement; copyright and legal notice.

13.1 Service & Calibration screen (service providers)

Figure 13-1 — TO DO: screenshot of the Service & Calibration screen

Open it from *Config* → *Service*, or by tapping the CAL DUE badge. Anyone can read it; nothing on it changes without the **service PIN**.

Control	What it does
Unlock / Lock	Enter the service PIN to unlock. Five wrong PINs lock the keypad for a minute.
Mark calibrated	Records today as the last calibration, clears CAL DUE and starts the next interval.
Interval line	Tap the *Interval* line in the card to set the days between calibrations: 90 is 3 months, 180 is 6 months, 0 turns the reminder off.
Clear zero	Sets the zero offset back to 0, so the Relative reading equals the Absolute reading again. Asks first; recorded in the service log.
Pump	Opens the Pump screen. Does not need the PIN.
Change PIN	Sets a new service PIN, 4 to 6 digits.
Factory reset	Restores every setting to factory values — including the input and pump calibration and the drop size — and marks calibration due. The clock, the service PIN and the interval are kept.

Table 13-1 Service screen controls

The same PIN protects the calibration settings elsewhere: *Input @ 4 mA* and *Input @ 20 mA* (*Config* → *Sensor*), and *Drop uL*, *uL/step* and Enter mL on the Pump screen. Once unlocked, the instrument stays unlocked for 5 minutes after the last service action while you move between the Service screen, Config and the Pump screen; going to Home, Titrate, Scope or Files locks it at once. A power interruption always leaves it locked.

Each service action is recorded in a service log on the instrument (on the USB stick instead, if the instrument's run store is not available). Export all copies it to the stick as `PCA_service_log.csv`. The PIN itself is never recorded.

Before you call

Please have ready: the firmware version (the last line of *Config*), a description of the symptom, and the exported file of an affected run. A photograph of the screen is often the fastest way to describe what you are seeing.